

Phoenix SecureCore Tiano™



User's Guide

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Document

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CHAPTER 1

INTRODUCTION

1.1 Introducing Phoenix SecureCore Tiano™

Phoenix SecureCore Tiano (SCT) is a new, advanced implementation of the industry-standard Tiano architecture that is UEFI compatible and is the finest implementation of Tiano in the world. This product delivers the best bring-up experience, the best customization experience, the fastest boot time, the highest reliability, full alignment with Intel's Green H, and a platform for innovation for third parties

The BIOS in a typical PC provides the same standard user experience for the systems it runs on. This allows a wide range of users to intuitively use any notebook computer, regardless of brand, for example. SecureCore Tiano can be configured to emulate the PC desktop or server experience; however, this is only one approach to interacting with the user, and it may not be the most appropriate one for a more specialized product. With over 1,000 ways of being configured by the OEM, it enables differentiation of products from competing products in quick and cost effective way.

This operating guide was written with the intent to describe all the user-visible features in Phoenix SecureCore Tiano that an OEM customer could enable and configure in their adaptation. However, the OEM may have chosen to disable any of these features, or configure them as necessary to fit a given application. Therefore, some of the features discussed in this operating guide may operate differently or not apply to your system.

1.2 About this Manual

This manual contains information important to users of systems that incorporate SecureCore Tiano; it describes how users interact with and configure the BIOS.

This manual does not contain programming information about BIOS interrupts or information for OEMs to adapt the BIOS to a given system; OEM customers are referred to the other applicable Phoenix Technologies' documentation that is made available with the SecureCore Tiano SDK.

The following is a chapter summary of this manual, for quick reference:

Chapter 1 – Introduction	Introducing this manual, SecureCore Tiano and its features.
Chapter 2 – POST	Learn about Power-On Self-Test (POST), including its user interfaces.
Chapter 3 – Setup	Configure the system with the System Setup Utility.
Chapter 4 – USB	Use USB keyboards, mice, hubs, and disks.
Chapter 5 – Security	Use the system’s security features.

1.3 SecureCore Tiano Features

To be compatible with today’s PC architecture, a BIOS must offer all the external software interfaces and certain internal nuances of the first IBM PC, IBM PC/XT, and IBM PC/AT BIOS implementations, as well as support the industry initiatives and operating systems that were introduced since those early days of the Personal Computer. SecureCore Tiano does this and more, building on a decade of support for these initiatives on the widest possible range of chipsets, CPUs, and board designs. Here’s a brief list of the features supported by SecureCore Tiano (and again, these features may be disabled or configured to any degree by the OEM):

- IA32 Support
- X64 Support
- Green H Compatibility
- ACPI Support
- HII 2.1
- CSM
- Native UEFI Booting Support
- Phoenix Native AHCI Bus Driver
- ATA Pass-Thru (IDE, PATA, SATA)
- SMBIOS 2.6
- Multi-Language Support (Build Tools, Fonts, Runtime)
- SLP 2.0 and 2.1
- Diagnostic Summary Screen During POST
- OEM Logo/Badging
- Security (TCG Callbacks, PBA)
- MultiBoot XP and MultiBoot 3 Boot Menus
- UEFI Boot Disable and Legacy Boot Disable
- Ibex Peak USB Dongle Driver (Phoenix)
- Legacy BIOS - Move BDA from platform to core
- Crisis Recovery
- IDE Bus Status Code Updates
- OEM RTC
- USB
- Full-featured POST time less than 5 seconds to pass MS Velocity test on all platforms.
- Versioning of Modules and Projects
- Preprocessor Support for ASL
- Centralized Build-Time Configuration
- Build projects in VS.NET 2005 and 2008 Toolchain
- Build in DOS box under these OSes:
 - o Windows XP
 - o Windows Vista
 - o Windows 7

- On-Target Operating Systems:
 - o MS Windows7 32-bit/64-bit Legacy
 - o MS Windows7 64-bit UEFI
 - o MS Windows Vista/SP1 64-bit UEFI
 - o MS Windows Vista 64-bit Legacy
 - o MS Windows Vista 32 bit - Install and boot
 - o MS Windows XP/SP3 32 bit Legacy only
 - o MS DOS 6.22
 - o Moblin 2.1
 - o Ubuntu 8.10 for Desktop
 - o Redhat 5.2 Linux For Desktop
 - o Fedora Core 10 for Desktop
- Foundation Features
 - o Configurable File Checksums
 - o Configurable PPI Limit
- Kernel Features
 - o Configuration Manager
 - o Trusted Computing Base (TCB)
 - o Integrated Security Authority (SA)
 - o Configurable Instrumentation
 - o Kernel Registry Configuration
 - o C Library
 - o Hardware Abstraction Layer
 - o X64\HAL.asm
 - o MicroKernel
 - o I/O System (File and Device Objects)
 - o MD5 Hash API
 - o System Information API
 - o List Management API
 - o Execution Context Object
 - o Bus Object
 - o Debug Object
 - o DPC Object
 - o Event Object
 - o Exception Object
 - o Kernel Applications
- Dynamic Linking for DXE drivers
- Access to Kernel Services and Objects
- DXE Execution Context

CHAPTER 2

POWER ON SELF TEST

2.1 About POST

When your system is powered on, SecureCore Tiano tests and initializes the hardware and programs the chipset and other peripheral components. This phase is called Power-On Self-Test, or POST for short. This chapter describes POST and how to interact with it as a user. Configuration of POST is described in detail in the System Setup Utility chapter.

2.2 POST Codes

POST is the process that configures the system to provide computing facilities to the user. During POST, thousands of CPU, chipset, Super I/O, and device registers must be configured and inspected with precise protocols. When the hardware fails to respond, POST may not be able to continue; for example, if the memory controller cannot be configured, the system cannot continue to initialize the display because the BIOS-level display driver (known as the Video BIOS, supplied by the silicon vendor) requires memory to work properly. Therefore there are cases where POST cannot continue, but has not performed all of the steps necessary to configure the user interface (keyboard and screen). SecureCore Tiano provides support for these early POST failure conditions.

During POST, 2-digit, hexadecimal-encoded values are written to an I/O port at address 80h. This I/O port may be monitored by a special display on the motherboard, or by a plug-in PCI card. If the system hangs during POST then these codes can be an effective way of determining the last action taken by POST. You can use these codes to look up a potential reason for the system hang in the *SecureCore Tiano Technical Reference Manual Volume 2 – Build*. Please note that while SCT POST codes are listed in this document, other components in the system, such as memory reference code, add-in network cards, video option ROMs, and operating systems, may also output their own codes.

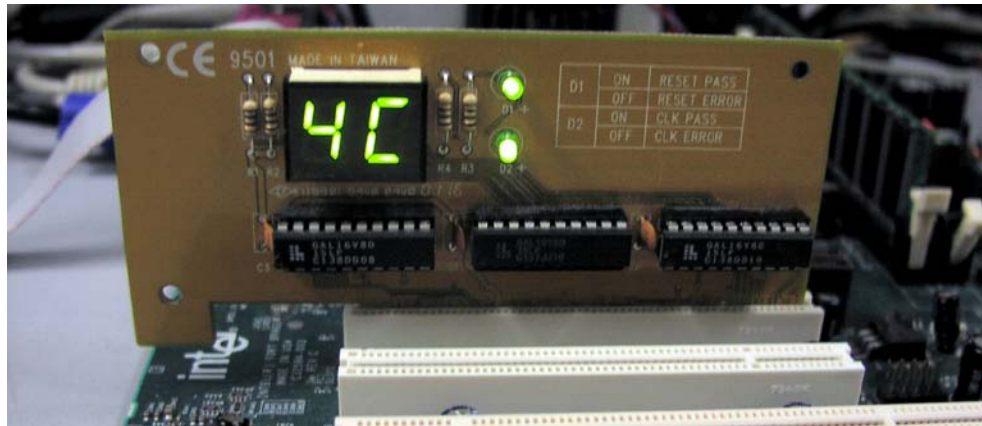


Figure 2.1. Using a PCI Port 80h Monitor Card.

2.3 The POST User Interface

Depending on the kind of equipment being initialized by SecureCore Tiano, POST may use a variety of components to interact with the user. These components are configurable from the pre-boot System Setup Utility (Setup).

2.3.1 Standard PC Keyboard and Video Display

POST normally interacts with the user through the standard video display (which might be connected to the system via VGA, AGP, TV, or other connections), and a PC keyboard, which may be connected to the system's legacy PS/2 keyboard port, or a USB port, depending on how the OEM has configured the BIOS to support these ports. See Figure 2.2 for a picture of a typical motherboard's connectors. The PS/2 keyboard and mouse connectors are stacked (keyboard usually color-coded in purple on the bottom) on the left-hand side. Along the bottom are the serial port DB9 male connector and the 15-pin VGA video connector, respectively. Above those two D-shell connectors is the 25-pin female parallel port connector. To the right are USB ports with network connector on top, and to the far right, the AC97 audio connectors.

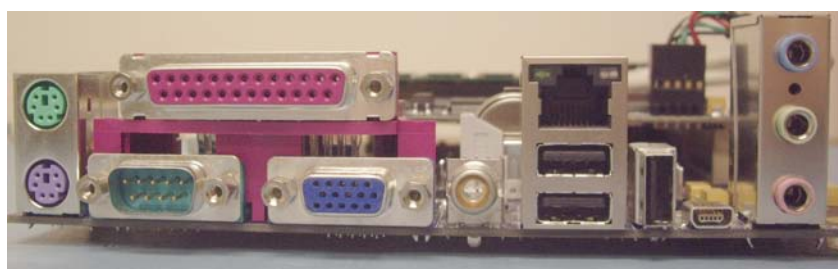


Figure 2.2. Connectors for PS/2 keyboard/mouse, video, serial port, parallel ports, and USB.

When the PC keyboard and video display is used, POST can be configured to have a graphical feel (with a splash screen and icons taking the place of the traditional PC BIOS's scrolling text.) This graphical user interface supports animation and sound to the extent that the OEM has configured it.

When operating in non-graphical mode, the video display is used to interact with POST in a classic way, meaning that POST displays white text on a black screen, such as the name of the system together with copyrights, memory test count-up, PnP and PCI configurations, and so on, scrolling as necessary (these displays can become lengthy with complex systems.)

2.3.2 Console Redirection

POST has a feature called Console Redirection, which redirects the character-based POST display to a serial port. The user connects a null modem serial cable between the system (the “target” system) and another computer system (the “host” system) which is using a terminal emulation program such as Windows HyperTerminal, TeraTerm, DOS Navigator, Telix, PROCOMM, or any other program to capture the console redirection data from the serial port.

With Console Redirection, the host’s keyboard is used to send keystrokes to the target system through the terminal emulation program over the serial cable; the target system receives these keystrokes as though they came from the PS/2 or USB keyboard connected directly to the system.

2.3.3 Headless Operation

POST can also be configured to not display any output, and therefore can operate systems that do not have a video display, keyboard, or serial port. This mode of operation is appropriate for systems such as communications equipment running in a data center.

2.4 POST User Intervention

With no typing on the part of the user, POST performs the activities as configured in the POST setup screen menu, and then begins processing the boot actions (typically, booting the operating system).

While POST ignores unrecognized keystrokes, certain keystrokes can direct POST to perform special functions. The OEM can edit, add or remove these typical key assignments:

- B / ^B keys – Enter the BIOS Boot Specification (BBS) Boot Menu, allowing interactive selection of boot action to be taken, rather than the set of boot actions configured in the BOOT Setup menu.
- D / ^D keys – Skip further disk drive detection, allowing POST to bypass waiting for drives to become ready when they have physically been disconnected from the system.
- ESC key – Skip further memory testing, allowing POST to progress quicker to its next step, which often is booting the OS.

2.5 Configuring POST

When the system is powered on for the first time, you may need to configure the system through the System Setup Utility (described in a later chapter) before peripherals, such as disk drives, are recognized by the BIOS. The information is written to non-volatile storage (sometimes Flash, or battery-backed CMOS RAM.)

The System Setup Utility allows the user to configure desired POST startup activities. The most important menus to consider when configuring POST are given below. See the System Setup Utility chapter for details about how to configure each item.

- POST Menu
 - Configure memory test strength and whether the memory test clears memory.
 - Request the system to pause if POST finds errors so the user can see the messages.
 - Enable or disable POST messages by category, such as PnP or PCI tables.
 - Perform automated rebooting for factory burn-in Q/A cycles.
- BOOT Menu
 - Build list of boot devices for POST to configure and attempt to boot from.

- o Select floppy drive hardware types (1.44MB, 720KB, 1.2MB, 360KB, etc.)
- o Select PATA drive cable types (40-pin, 80-pin, autodetection.)
- o Select PATA UDMA modes authorized by user.

2.6 Booting Operating Systems

For the most part, booting a desktop or server operating system on a system using SecureCore Tiano works the same as on any other desktop or server system. The system scans down its BBS boot list, looking for activities to perform, and attempts to perform them in the specified order.

As explained in Chapter 3, on the Boot menu of the System Setup Utility, there are two types of items on this BBS list; boot activities and drives. Boot activities consist of items like “Boot Windows CE”, “Boot EFI”, “Boot from PCI Slot 2”, “Boot Debugger”, “Boot Setup”, and so on. Drives are storage devices from which to load operating systems. It is customary to run boot activities before drives.

When a boot activity is performed, the activity may involve reading a file (i.e., Boot Windows CE requires that NK.BIN be loaded from a disk drive, and Boot EFI requires that EFILDR.BIN be loaded from a disk drive.) In these cases, the file to be loaded must reside in the root directory of a FAT file system located on one of the drives specified in the BBS list.

When the boot action is a drive, POST attempts to boot the Master Boot Record (MBR) or Partition Boot Record for a floppy, residing on the disk.

Here are two examples of how activities and drives can be combined.

Suppose that one hard drive is attached to a system and the user wants to try booting Windows CE, then EFI, before a legacy operating system (Windows XP) from the MBR. To accomplish this, the BBS list would look like the following:

1. Boot Windows CE (searches all disks in specified order for NK.BIN)
2. Boot EFI (again, searches all disks in specified order for EFILDR.BIN)
3. IDE Primary Master

To boot a legacy operating system first, before attempting to boot Windows CE or EFI, the order would look like the following:

1. IDE Primary Master
2. Boot Windows CE
3. Boot EFI

CHAPTER 3

SYSTEM SETUP UTILITY

3.1 The System Setup Utility

The System Setup Utility (Setup) is highly configurable. Your version may not include all the features described in this chapter, or it may include additional OEM-proprietary features not documented here. However, it is very likely that your system includes the basic set of configuration options.

Setup may be run from either the target system keyboard and video display or from a terminal emulator program running on a host computer connected to the target system through a serial cable.

3.2 Setup Menus

The standard Setup menus are shown below in the order they generally appear in the menu:

Main	Display main system components and allow editing of date and time.
Advanced	Configure advanced settings such as language, boot configuration, ACPI, HDD, IMC, and more.
Security	Configure which BIOS features require user authentication before they perform their functions.
Boot	Configure boot actions and boot devices.
Exit	Save changes and exit, discard changes and exit, or restore factory default settings.

3.3 Navigating Setup Menus and Fields

Navigation (moving your cursor around, selecting items, and changing them) is easy in Setup. The following chart is a helpful user reference:

UP key (also ^E)	Move the cursor to the line above, scrolling the window as necessary.
DOWN key (also ^X)	Move the cursor to the line below, scrolling the window as necessary.
LEFT key	Go back to the menu to the left of the currently-displayed menu in the menu bar.
RIGHT key	Go forward to the menu to the right of the currently-displayed menu in the menu bar.
PGUP key	Move the cursor up several lines (a full window's worth), scrolling the window as necessary.
PGDN key	Move the cursor down several lines (a full window's worth), scrolling the window as necessary.
HOME key	Move the cursor to the first configurable field in the current menu, scrolling the window as necessary.
END key	Move the cursor to the last configurable field in the current menu, scrolling the window as necessary.
ESC key	Exit the System Setup, discarding all changes (except date/time changes, which take place on-the-fly.)
TAB key	Move the cursor down to the next configurable field.
Shift-TAB key (backtab)	Move the cursor up to the last configurable field.
+ key	Toggle an Enable/Disable field, or increase a numeric field's value.
- key	Toggle an Enable/Disable field, or decrease a numeric field's value.
SPACE key	Toggle an Enable/Disable field.
BKSP key	Reset an Enable/Disable or multiple-choice field, or back-up in numeric or string fields.
Digits (0-9)	Used to enter numeric parameters.
Alphabetic (A-Z, a-z)	Used to enter text data on ASCII fields such as email addresses.
Special symbols (!@#\$\$%^&* _ - += {} [], etc.)	Used to enter special text on ASCII fields that permit these characters.

The basic idea when using Setup is to navigate to the menus containing configuration fields, and change those fields as desired. When your settings are complete, navigate to the EXIT menu, and select "Save Settings and Restart". This causes the settings to be stored in nonvolatile memory before reboot, so that POST can configure itself with the new settings. After reboot it may be desirable to reenter Setup to adjust settings.

Once the system begins to boot, Setup cannot be entered. This is due to the fact that the memory used by the BIOS configuration manager is deallocated by the system BIOS, so that it can be used by the OS. To enter the Setup after boot you must power cycle your system.

3.4 Main Setup Menu

This menu, shown in Figure 3.1, is the default screen (unless disabled by the OEM).

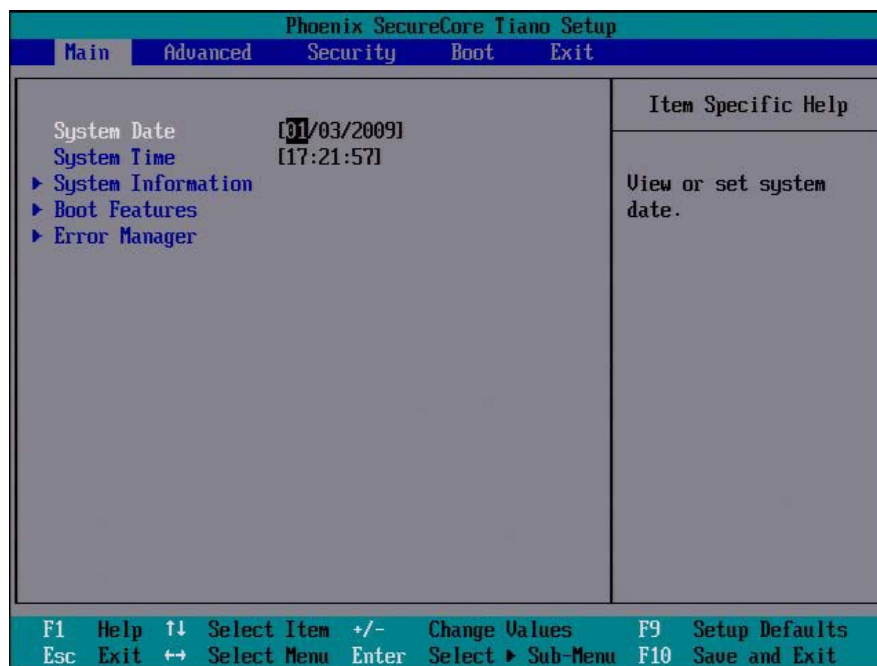


Figure 3.1. Main Setup Menu.

The Main screen shows the system date and time. The Real Time Clock fields are editable with keystrokes. To navigate through the MM/DD/YYYY and HH:MM:SS fields, use the TAB and BACKTAB (SHIFT + TAB) keys. The hours are normally specified in military time; thus 13 means 1pm, or one hour after noon, whereas 01 means 1am, or one hour after midnight. When the cursor leaves Real Time Clock (RTC) fields, they either affect the battery-backed RTC right away, allowing the system to continue with your new settings, or they revert back to old values if the new values are not valid entries.

The Main menu also provides access to the System Information submenu shown in section 3.4.1. This submenu shows information concerning the system BIOS, processor, memory, and other items configured by the OEM. The Phoenix System BIOS information is created when the BIOS is built; this information is useful when seeking system support.

PLEASE CALL YOUR OEM VENDOR FOR BIOS SUPPORT; DO NOT CALL PHOENIX TECHNOLOGIES DIRECTLY UNLESS YOU HAVE A DEVELOPMENT BOARD FROM A SILICON VENDOR, AND THE MATERIALS WITH THE BOARD DIRECT YOU TO CALL PHOENIX TECHNOLOGIES. Phoenix Technologies supports its OEM customers, and OEMs support their customers. Phoenix Technologies cannot support end users directly, not only for volume reasons, but also because Phoenix Technologies has security agreements with many of its OEM customers prohibiting such support. Additionally, OEMs often adapt SecureCore Tiano to such an extent that it is not possible for Phoenix Technologies to adequately provide support in a generic sense; it makes sense for the OEM to perform this role.

PHOENIX TECHNOLOGIES WELCOMES SUPPORT CALLS FROM CUSTOMERS EVALUATING SILICON VENDOR REFERENCE PLATFORMS CONTAINING A COPY OF PHOENIX TECHNOLOGIES' SecureCore Tiano BUILT BY PHOENIX TECHNOLOGIES SPECIFICALLY FOR USE WITH THAT REFERENCE DESIGN.

3.4.1 System Information

This submenu provides information about the system BIOS, CPU, and memory, as shown in Figure 3.2.

Phoenix SecureCore Tiano Setup		
Main		
System Information		
BIOS Version	[HuronRiver_CRBWW29	] 1
Build Time	[08/26/2010	] 1
Processor Type	[Genuine Intel(R) CPU 0 @ 1.80GHz	] 1
Processor Speed	[1.80GHz	] 1
System Memory Speed	[1067 MHz	] 1
L2 Cache RAM	[256 KB	] 1
Total Memory	[4096 MB	] 1
Memory Mode	[Memory MODE_DUAL_INTERLEAVE	] 1
Memory Channel A Slot 0	[2048 MB (DDRIII1067)	] 1
Memory Channel A Slot 1	[	] 1
Memory Channel B Slot 0	[2048 MB (DDRIII1067)	] 1
Memory Channel B Slot 1	[	] 1
F1 Help ↑↓ Select Item +/- Change Values F9 Setup Defaults Esc Exit ↔ Select Menu Enter Select ► Sub-Menu F10 Save and Exit		

Figure 3.2. System Information.

The CPU information is normally obtained directly from the CPU's Model-Specific Registers (MSRs).

The system memory information does not describe physical RAM; rather it describes the RAM as configured, subtracting RAM used for System Management Mode, Shadowing, Video buffers, and other uses. This provides realistic values about how much memory is actually available to operating systems and applications.

3.5 Advanced Menu

As shown in Figure 3.3, the Advanced menu contains configuration settings grouped by submenus.

Note: Setting items on the Advanced screen to incorrect values may cause the system to malfunction.

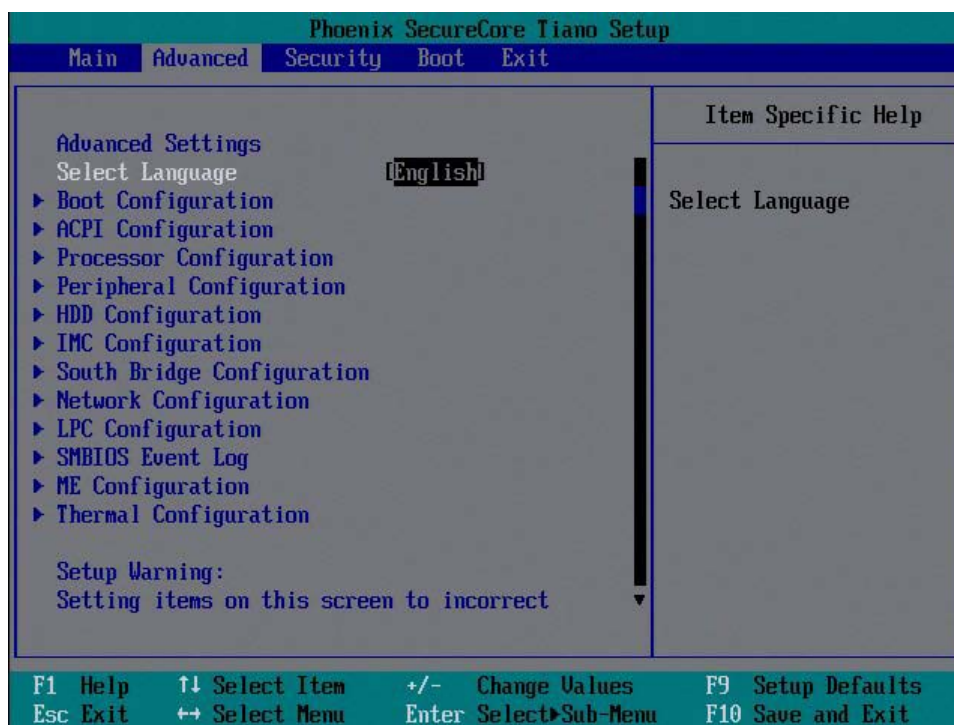


Figure 3.3. Advanced Setup Menu.

Select Language	Select the language for the setup screen text.
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The following table describes the settings associated with the Boot Configuration menu.

Quick Boot	Enables time-optimized POST, causing certain preconfigured OEM optimizations to be made when the system boots.
High Resolution Graphics	Enables high resolution graphics.
Diagnostic Splash Screen	Enables graphical POST, including animation, sound, icons, advertisements, and other multimedia objects that may be configured by the OEM.
Diagnostic Summary Screen	Enables the diagnostic summary screen.
BIOS Level USB	Enable USB at the BIOS level.
USB Legacy	Enables support for USB devices including mouse, keyboard, mass storage, and so on.
Console Redirection	Enables console redirection over a serial port. Enabled – causes the BIOS to always use the serial port as the console, without testing for the presence of the terminal emulation program. Disabled – causes the BIOS to never invoke console redirection, but instead always use the main keyboard and video display. If there is no keyboard or video display, the system operates headless.
UEFI Boot	Enables the Unified Extensible Firmware Interface.
Legacy Boot	Enable this option to bypass some drivers and speed up POST.
Boot Priority	Set the priority of boot options.

3.5.1 ACPI Configuration

The following table describes the settings associated with the ACPI Configuration menu.

FACP - RTC S4 Flag Value	Controls the value for the RTC S4 flag in the FACP table. Enabled – the RTC alarm can wake the system from the S4 state. Disabled – the RTC alarm can wake the system from S1, S2, or S3 sleep states.
APIC - IO APIC Mode	Enable/Disable the Advanced Programmable Interrupt Controller (APIC) – IO APIC Mode.
ALS Support	Specify the type of ACPI support. Legacy – ALS support through the IGD INT10 function. ACPI – ALS support through an ACPI ALS driver.
EMA Support	Enables the EMA device in an ACPI environment.
MEF Support	Enables Mobile East Fork (MEF) support in an ACPI environment.
Enabled PTID	Enables power and temperature instrumentation details.
FACP – PM Timer Flag Value	Enables the PM timer Flag in the FACP table. Enabled – indicates that the timer value is implemented as a 32-bit value. Disabled – indicates the timer value is implemented as a 24-bit value.

3.5.2 Processor Configuration

The following table describes the settings associated with the Processor Configuration menu.

Active Processor Cores	Number of cores to enable in each processor package.
Intel® HT Technology	Enables Intel® Hyper-Threading (HT) Technology.
Dynamic FSB Switching	Enables processor Dynamic FSB Frequency Switching (Bus GV.)
Enabled XD	Enables memory segregation for improved security.
Machine Check	Enables machine check exception handling.
Fast Strings	Enables processor fast strings.
Intel® Virtualization Technology	Enable this to consolidate multiple environments into a single server, workstation, or PC.
► Processor Power Management	Press Enter to access this submenu (see section 3.5.3.1)

3.5.2.1 Processor Power Management

The following table describes the settings associated with the Processor Power Management menu.

Intel® SpeedStep™	Enables processor performance states (P-States).
Boot Performance Mode	Set the default performance state at startup. Select either Max Performance, Max Battery, or Auto.

Turbo Mode	Enables Turbo mode. When enabled, specify values for the Long Power Limit, Long Power Limit Time, and Short Power Limit.
C-States	Enables power saving modes.
Extend C-States	Enables extended power saving modes.
C3-State	Enables Sleep (C3) state
C6-State	Enables Deep Power Down (C6) state.
C7-State	Enables Deeper Sleep (C7) state.
C7s-State	Enables C7s state.
C-State Auto Demotion	Enable auto demotion of C states.

3.5.3 Peripheral Configuration

The following table describes the settings associated with the Peripheral menu.

Spread Spectrum Clock	Enables the clock chip Spread Spectrum feature.
PCIe SR-IOV Support	Enables support for Single Root I/O Virtualization which enables efficient sharing of a single I/O device among multiple virtual machines.

3.5.4 HDD Configuration

The following table describes the settings associated with the HDD Configuration menu.

HDD Unlock	Enables the HDD password unlock in the OS.
LED Locate	Enables LED in the OS.

3.5.5 Memory Configuration

DIMM Profile	Specifies the SPD XMP profile for XMP-supported DIMM.
Memory Frequency	Specifies the memory frequency.
Max TOLUD	Specify the maximum Top of Low Usable DRAM (TOLUD). Available options include: 3.25 GB 3.00 GB 2.75 GB 2.50 GB 2.25 GB 2.00 GB
OneN Mode Support	Enable OneN support.
Scrambler Support	Enables scrambler support.

3.5.6 System Agent (SA) Configuration

DMI Settings	Configure DMI settings
Intel® VT for Directed I/O (VT-d)	Specifies configuration settings for Intel Virtualization Technology for Directed I/O.
Graphics Configuration	Specifies graphic settings.
PEG Configuration	Specifies PCI Express Port Configuration.
Debug Align	Enables Debug Align control for DMI and PEG.
DeEmphasis	Configures DeEmphasis control for PEG. Options include: -6 dB -3.5 dB

3.5.6.1 DMI Settings

DMI Vc1 Control	Enables DMI Virtual channel 1 control.
DMI Vcp Control	Enables DMI Virtual channel P control.
DMI Vcm Control	Enables DMI Virtual channel M control.
DMI Extended Sync	Enables DMI Extended Sync Control.
DMI Gen2 Support Control	Enables DMI Gen2 Support Control.

3.5.6.2 Intel(R) VT for Directed I/O (VT-d) Settings

Intel(R) VT for Directed I/O (VT-d)	Enables Intel(R) Virtualization Technology for Directed I/O (VT-d) by reporting the I/O device assignment to VMM through DMAR ACPI Tables.
Queued Invalidation	Enables Queued Invalidation for Intel(R) Virtualization Technology for Directed I/O (VT-d).
Interrupt Remapping	Enable/Disable Interrupt Remapping for Intel(R) Virtualization Technology for Directed I/O (VT-d).

3.5.6.3 Graphics Settings

Internal Graphics	Enable/Disable the Internal Graphics Device. This has no effect if external graphics are present.
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3.5.6.4 PEG Settings

PCI Express Port Link Width Settings	Set the PCI Express link width of configurable ports. Specify one of the following: X4,X4 X4,X8 X8,X4 X8,X8 X16
PCI Express ASPM Support	Control PCI Express Active State Power Management settings.
PCI Express ASPM L1 Support	Control PCI Express ASPM L1
PCI Express URE	Control PCI Express Unsupported Request Reporting Enabled.
PCI Express FEE	Control PCI Express Fatal Error Reporting Enabled.
PCI Express NFE	Control PCI Express Non-Fatal Error Reporting Enabled.
PCI Express CEE	Control PCI Express Correctable Error Reporting Enabled.
PCI Express SFE	Control PCI Express System Error on Fatal Error Enabled.
PCI Express SNE	Control PCI Express System Error on Non-Fatal Error Enabled.
PCI Express SCE	Control PCI Express System Error on Correctable Error Enabled.
PCI Express Completion Timeout	Control PCI Express Completion Timeout Enabled.
PCI Express PME Interrupt	Control PCI Express System Error on Correctable Error Enabled.
PCI SERR#	Control PCI SError Enable (SERR#)
PCI PERR#	Control PCI PError Enabled.
PEG1 (Device 5) Settings	Control PCI Express Active State Power Management settings.

3.5.7 South Bridge Configuration

The following table describes the settings associated with the South Bridge Configuration menu.

SMBUS Device	Enables SMBUS Device.
Port 80h Cycles	Specifies whether the Port 80h cycles are sent to LPC Bus or PCI Bus.
PCI Clock Run Logic	Enables the PCIe clock gate power saving feature.
Azalia	Enables Azalia device detection.
EHCI1/2	Enables the EHCI Firmware Technology driver, allowing USB Boot feature to use high speed transfers on USB 2.0 ports in the system.
USB Ports Per-Port Disabled Control	Enables control of each USB port.
► SB PCI Express Config	Press Enter to access this submenu (see section 3.5.7.1)

3.5.7.1 SB PCI Express Configuration

The following table describes the settings associated with the South Bridge PCI Express Configuration menu.

PCI Express Root Port Clock Gating	Enables PCI Express root port clock gating.
PCI Express Controller DMI Setting	Configures the DMI settings of the PCI Express controller. Options include Disabled; L0S; L1; L0S and L1; and Auto.
► PCI Express Port (1-8) Config	Press Enter to access these submenus (see section 3.5.7.1.1). PCI Express Port 6 is dedicated to LAN.

3.5.7.1.1 PCI Express Port Configuration

The following table describes the settings associated with the PCI Express Root Port menu for Ports 1-5, 7, and 8. (Note that PCI Express Port 6 is dedicated to LAN.)

PCI Express Root Port (1-8)	Enables the PCI Express root port.
ASPM	Specifies Active State Power Management (ASPM) settings. Options include Disabled; L0S; L1; L0S and L1; and Auto.
HOT PLUG	Enables hot plug support.
URR	Enables unsupported request reporting.
FER	Enables fatal error reporting.
NFER	Enables non-fatal error reporting.
CER	Enables correctable error reporting
CTD	Enables completion timer timeout support.
SEFE	Enables System Error on Fatal Error (SEFE) support.
SENF	Enables System Error on Non-Fatal Error (SENF) support.
SECE	Enables System Error on Correctable Error (SEC) support.
PME Interrupt	Enables PME Interrupts.
PME SCI	Enables PME SCI support.

3.5.8 Network Configuration

The following table describes the settings associated with the Network Configuration menu.

PCH Internal LAN	Enables PCH Internal LAN.
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LAN OPROM Selection	Enables support for LAN OPROM, which is used for quick boot minimal configuration.
Wake on LAN	Enables the PCH Internal Wake on LAN capability.
ASF Support	Enables alert specification format support.

3.5.9 LPC Configuration

The following table describes the settings associated with the Low Pin Count (LPC) Configuration menu.

Onboard UART1	This option controls the Onboard UART1 Address. When enabled, UART1 uses address 0x3F8h and IRQ4.
Onboard CIR (UART2)	This option controls the Onboard CIR address. When enabled, CIR uses UART2 address 0x2F8h and IRQ 3.

3.5.10 SMBIOS Event Log

The following table describes the settings associated with the SMBIOS Event Log Configuration menu.

Event Log Validity	Read-only field.
Event Log Capacity	Read-only field.
Event Log	Enables the Event Log.
View SMBIOS event log	Displays a pop-up window showing SMBIOS events.
Mark SMBIOS events as read	Marks all SMBIOS events as read.
Clears SMBIOS events	Clears all SMBIOS events from the log.

3.5.11 ME Configuration

The following table describes the settings associated with the Management Engine Configuration menu.

Intel ME	Enables the Intel Management Engine.
ME FW Version	Read-only field.
ME Firmware	Read-only field.
End of POST Message	Enables an End of POST message, which will be sent to the ME.

3.5.12 Thermal Configuration

The following table describes the settings associated with the Thermal Control Configuration menu.

CPU Thermal Configuration	Opens a submenu for processor thermal management. (See section 3.5.12.1.)
Intelligent Power Sharing	Opens a submenu for intelligent power sharing configuration. (See section 3.5.12.2.)
Platform Thermal Configuration	Opens a submenu for platform thermal management. (See section 3.5.12.3.)

3.5.12.1 CPU Thermal Configuration

The following table describes the settings associated with the Processor Thermal Management menu.

TM2	Enables processor Thermal Monitor 2 (TM2) thermal control. Note: TM2 requires GV3.
TM1	Enables processor Thermal Monitor (TM1) thermal control.
EMTTM	Enables processor Enhanced Multi-Threaded Thermal Monitoring. Note: EMTTM requires GV3.
Turbo Mode	Enables processor Turbo Mode. Note: EMTTM must be enabled in order to enable Turbo Mode.
Bi-directional PROCHOT#	The PROCHOT# is driven when a processor thermal sensor trips. If this option is enabled, then external agents can drive PROCHOT# to throttle the processor.
Thermal Offset	Enables the KSC to use Thermal Offset (read from CPU MSR 03Fh) to adjust thermal management.
DTS	Enables the CPU digital thermal sensor function.
DTS Calibration	Enables the digital thermal sensor calibration function for thermal management rather than processor defaults.

3.5.12.2 Intelligent Power Sharing

The following table describes the settings associated with the Intelligent Power Sharing menu.

Intelligent Power Sharing	Enables Intelligent Power Sharing configuration menu. Note: DTS must be enabled for Power Sharing to function.
CPU Turbo	Enables CPU turbo.
MCH Turbo	Enables MCH turbo.
PPEC Config	Specify the Process Power Error correction value.
IPS Policy	Specify the platform BIOS policy preference. Default – BIOS does not have an overriding preference for allocation of the Turbo power budget, allowing the driver to dynamically determine the preference at run-time. (Recommended) Processor Core – BIOS has an overriding preference for the processor component to receive an increased portion of the Turbo power budget. Balanced – BIOS has an overriding preference for the processor and graphics components to receive fairly shared portions of the Turbo power budget. Graphics – BIOS has an overriding preference for the graphics component to receive an increased portion of the Turbo power budget.
Core Temp Limit	Enables the Core Temperature Limit feature.
Core Temp Limit Value	Specify a Core Temperature Limit value.
MCH Power Limit	Enables the MCH power clamp.
MCH Power Limit Value	Specify the maximum MCH power clamp value in tenths of a watt. The maximum value is 511.
Processor Power Limit	Enables the maximum processor power clamp.

Processor Power Limit Value	Specify the maximum processor power clamp value in tenths of a watt. The maximum value is 1023.
Core Power Limit	Enables the maximum core power clamp.
Core Power Limit Value	Specify the maximum core power clamp value in tenths of a watt. The maximum value is 1023.
Run Time Interface	Enables selection of a runtime interface for PCH communication.

3.5.12.3 Platform Thermal Configuration

The following table describes the settings associated with the Platform Thermal Configuration menu.

Thermal Data Report Enable	Enables Thermal Data reporting.
MCH Temp Read Enable	Enables temperature reading on MCH.
PCH Temp Read Enable	Enables temperature reading on PCH.
CPU Energy Read Enable	Enables CPU energy reading.
CPU Temp Read Enable	Enables temperature reading on the CPU.
TS On Dimm Enable	Enables temperature reading in DIMMs.
DIMM1 Temp Read Enable	Enables temperature reading on DIMM1.
DIMM2 Temp Read Enable	Enables temperature reading on DIMM2.
DIMM3 Temp Read Enable	Enables temperature reading on DIMM3.
DIMM4 Temp Read Enable	Enables temperature reading on DIMM4.
Alert Enable Lock	Enables locking of the Alert Enable State for Proc, DIMMs, PCH, and MCU/MCH.
DIMM Alert Enable	Enables assertion of the PCH temperature alert pin if the temperature of DIMMs1-4 exceeds the limits.
PCH Alert Enable	Enables assertion of the PCH temperature alert pin if the PCH temperature exceeds the limit.
MCH Alert Enable	Enables assertion of the PCH temperature alert pin if the temperature of MCU/MCH/Graphics exceeds the limits.
CPU Alert Enable	Enables assertion of the PCH temperature alert pin if the temperature of the processor core exceeds the limit.
ME SMBus Thermal Reporting	Enables ME SMBus thermal reporting configuration.
ME SMBus Buffer Length	Specifies the SMBus block read message length for EC.
Thermal Reporting EC PEC	Enables packet error checking (PEC) for SMBus block read.
Select slots with TS on DIMM's	Specify temperature report for slots with TS on DIMM.

3.6 Security Setup Menu

The Security Setup menu enables the administrative user to select which BIOS features require user authorization to operate. Figure 3.4 shows a typical Security Setup menu, which is OEM-configurable.

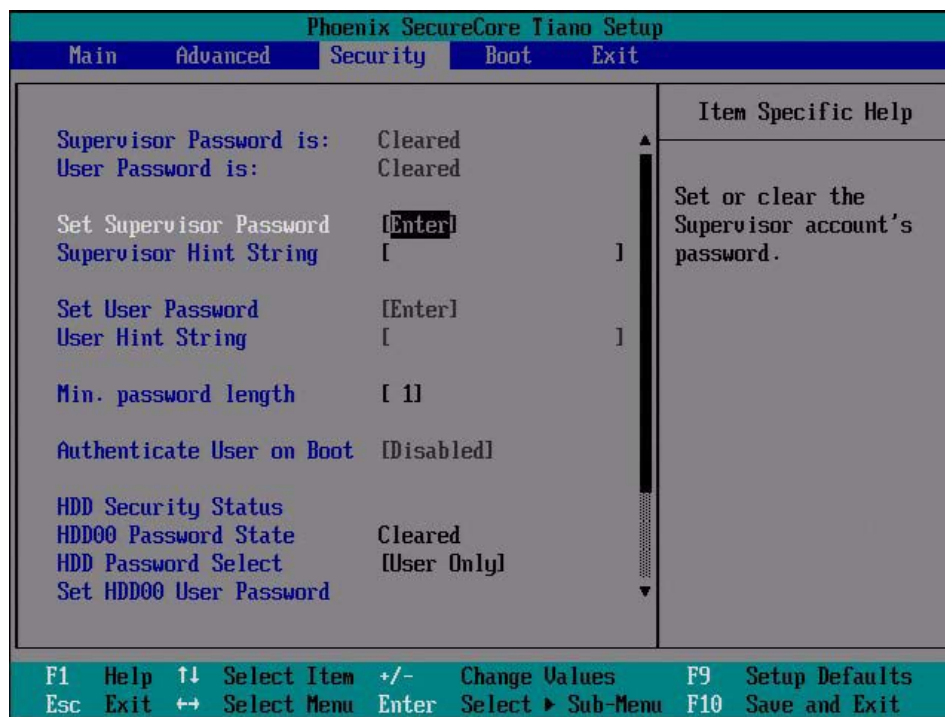


Figure 3.4. Security Setup Menu.

The following table presents the configurable features in the Security Setup menu. Note the picture does not necessarily include all of the fields documented here.

User Password is	Specify whether the User Password is set or cleared.
Supervisor Password is	Specify whether the Supervisor Password is set or cleared.
Set Supervisor Password	Press Enter to specify a supervisor password.
Supervisor Hint String	Press Enter to specify a hint string for the supervisor password.
Set User Password	Press Enter to specify a user password.
Min. password length	Specify the minimum password length.
User Hint String	Press Enter to specify a hint string for the user password.
Authenticate User on Boot	Enables the user authentication prompt on boot.
HDD01 Password State	Specify whether the HDD user and master passwords are set or cleared.
HDD Password Select	Specify whether to enable User-only support for HDD or User and Master support.
Set HDD01 User Password	Specify and confirm the HDD user password.
Set HDD01 Master Password	Specify and confirm the HDD master password.
Disable HDD01 User Password By Master	Allows the Master to disable the HDD01 user password.
Trusted Platform Module (TPM)	Read-only field that shows whether a TPM devices is connected to the board.

Administration of users and authorized permissions is beyond the scope of this User's Guide. OEM customers should consult their SecureCore Tiano SDK documentation and supply documentation supporting their security administration platform for their users in the applicable data system administration document.

3.7 Boot Setup Menu

The Boot menu allows the system’s boot priority order and boot options to be configured. The main Boot menu is shown in Figure 3.5.

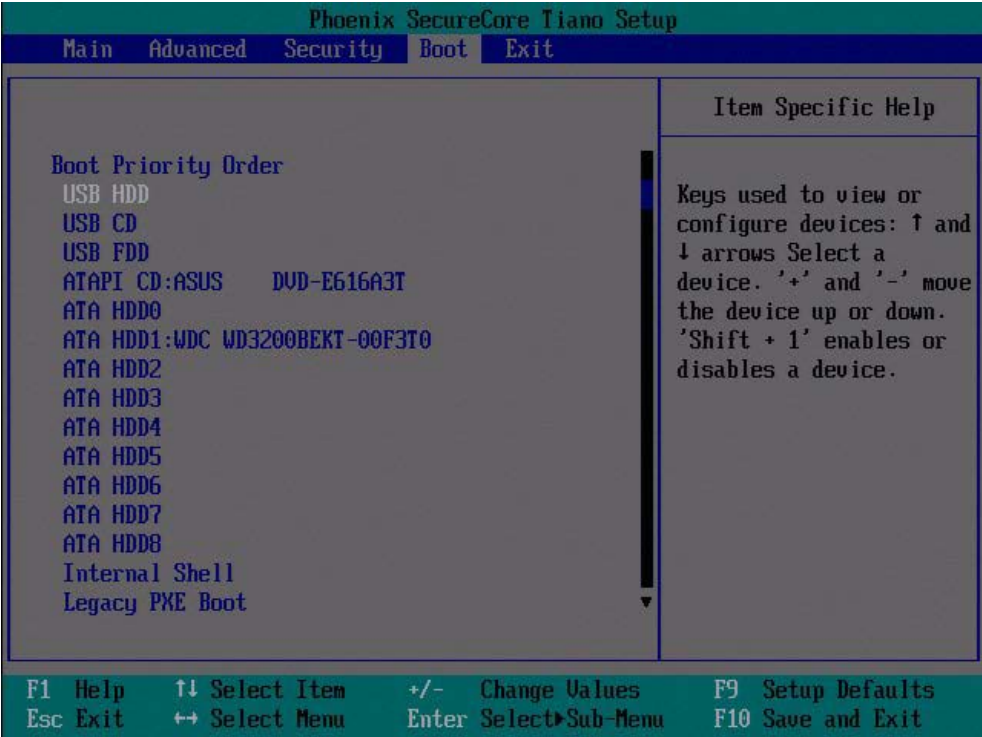


Figure 3.5. Boot Setup Menu.

The following table presents the configurable features in the Boot setup menu.

Boot Priority Order	Specify the boot priority order for the list of devices. Select a device by moving the ↑ or ↓ buttons. Once a device is selected, use the “+” or “-” buttons to move the device up or down in the priority order list. Finally, select Shift+1 to enable or disable a device.
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3.8 Exit Setup Menu

The Exit menu provides methods for saving, discarding, or reloading the settings within System Setup. This menu is shown in Figure 3.6.

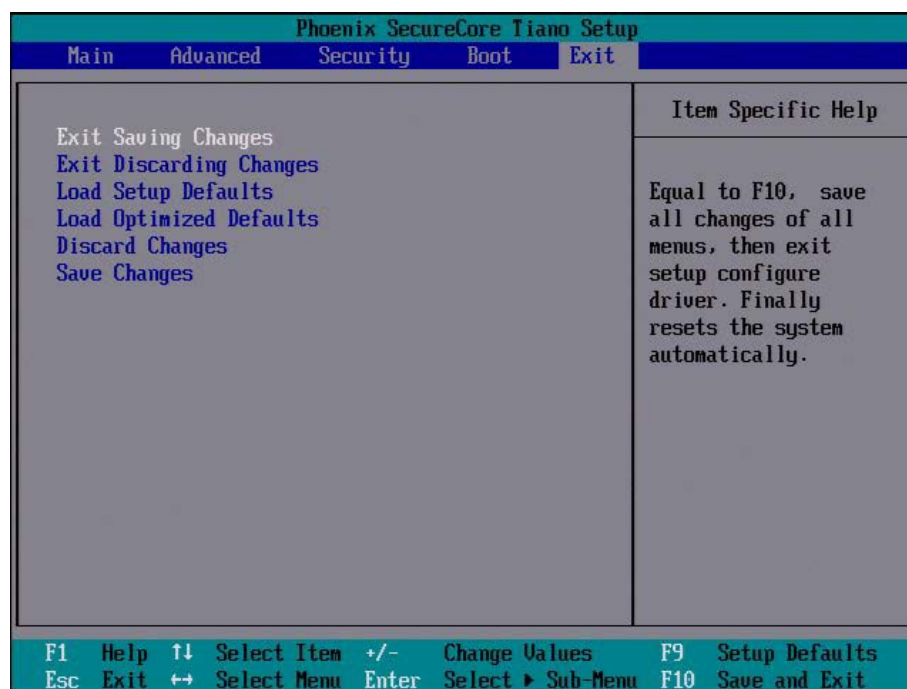


Figure 3.6. Exit Setup Menu.

The following table presents the configurable features in the Boot setup menu.

Exit Saving Changes	Selecting this is equivalent to pressing F10 . All changes on all menus will be saved, and the system will exit the setup menu and will reset automatically.
Exit Discarding Changes	Selecting this is equivalent to pressing Esc . No changes will be saved, and the system will exit the setup menu and will reset automatically.
Load Setup Defaults	Selecting this is equivalent to pressing F9 . The system will reset to the default configuration.
Load Optimized Defaults	Specify to load settings for optimized boot time and system performance.
Discard Changes	Selecting this causes the system to load the original options at this current boot time. The system will not exit the setup menu after discarding changes.
Save Changes	Selecting this causes the system to save all changes on all menus. The system will not exit the setup menu after saving changes.

To select any of these options, position the cursor over the option and press the **ENTER** key. Pressing the **ESC** key at any time within the System Setup is equivalent to choosing “Exit Discarding Changes.”

All of these options require verification before performing the selected action; otherwise, the system configuration might be saved or lost by accident. Figure 3.7 illustrates the verification popup for saving and exiting; the other options are similar.

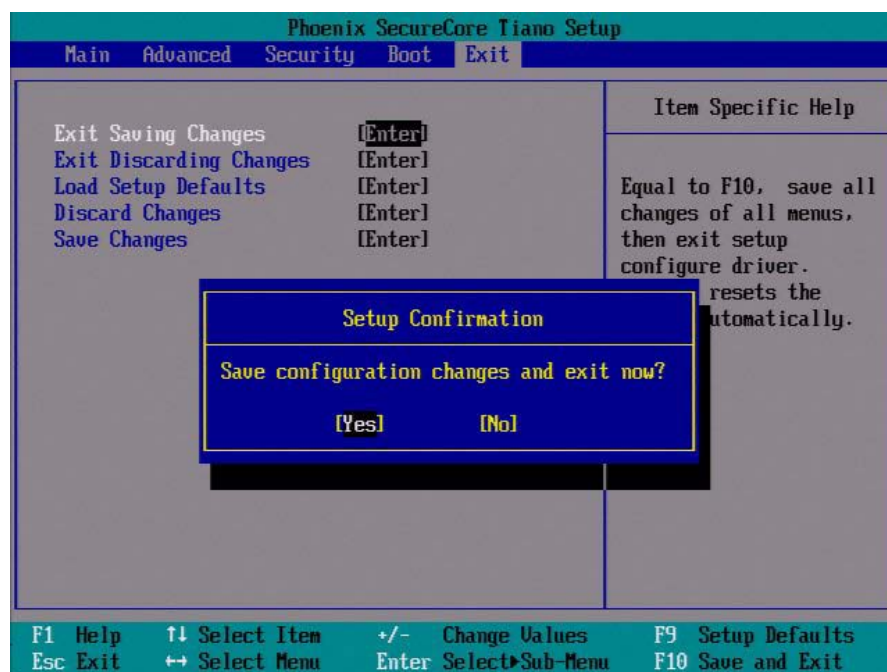


Figure 3.7. Popup requesting verification before exiting System Setup.

CHAPTER 4

UNIVERSAL SERIAL BUS

4.1 USB Features

The system BIOS provides comprehensive support for USB mass storage and human interface devices, including the ability to boot operating systems from USB devices, and emulate a PS/2 keyboard and mouse with their USB equivalents.

USB is actually a sophisticated hierarchical bus that supports multiple keyboards, mice, floppy disks, hard disks, CDROM/DVD/CDR and other devices, such as printers, scanners, cameras, and other equipment. Of these, only bootable mass storage devices are supported by the USB boot feature, and only those USB devices that support the industry standard USB Boot HID protocol are supported (virtually all keyboards and mice, but not strictly all.) In general, standard USB keyboards and mice purchased at a computer store will work with the BIOS USB feature.

The USB stack features the industry-standard handoff, as defined by Microsoft documentation, allowing the preboot environment to configure and use the USB devices up to exactly the point at which the OS wishes to take over the devices' operation. When the OS wishes to hand-back the devices, the USB stack follows the industry-standard handback protocol to resume control over the devices. Operating systems not following the industry-standard protocols for handoff and handback may have limitations when used with this full-featured USB stack.

Some operating systems, like DOS and Windows NT, do not have built-in USB device support, and so never execute a handoff or handback protocol; instead, these operating systems and their application programs believe they are communicating with real PS/2 devices and real "INT 13h" disk drives in the system.

The USB services support arbitrarily complex bus topologies incorporating USB hubs, up to eight USB keyboards and mice simultaneously, and a large variety of USB mass storage devices including floppy disk drives, hard disk drives, CD/DVD ROM drives, CompactFlash readers, Zip disk drives, DiskOnKey devices and much more.

WARNING: USB hubs and other devices may be powered or unpowered. It is up to the user to connect devices in such a configuration that each hub and device has sufficient power to operate properly. When this configuration is incorrect, devices may not have enough power to function, and as a result may operate erratically or not at all.

The Boot menu deals with bootable actions, and if configured by the OEM, can include bootable USB devices in the boot action list. This menu is shown in Figure 4.1 below.

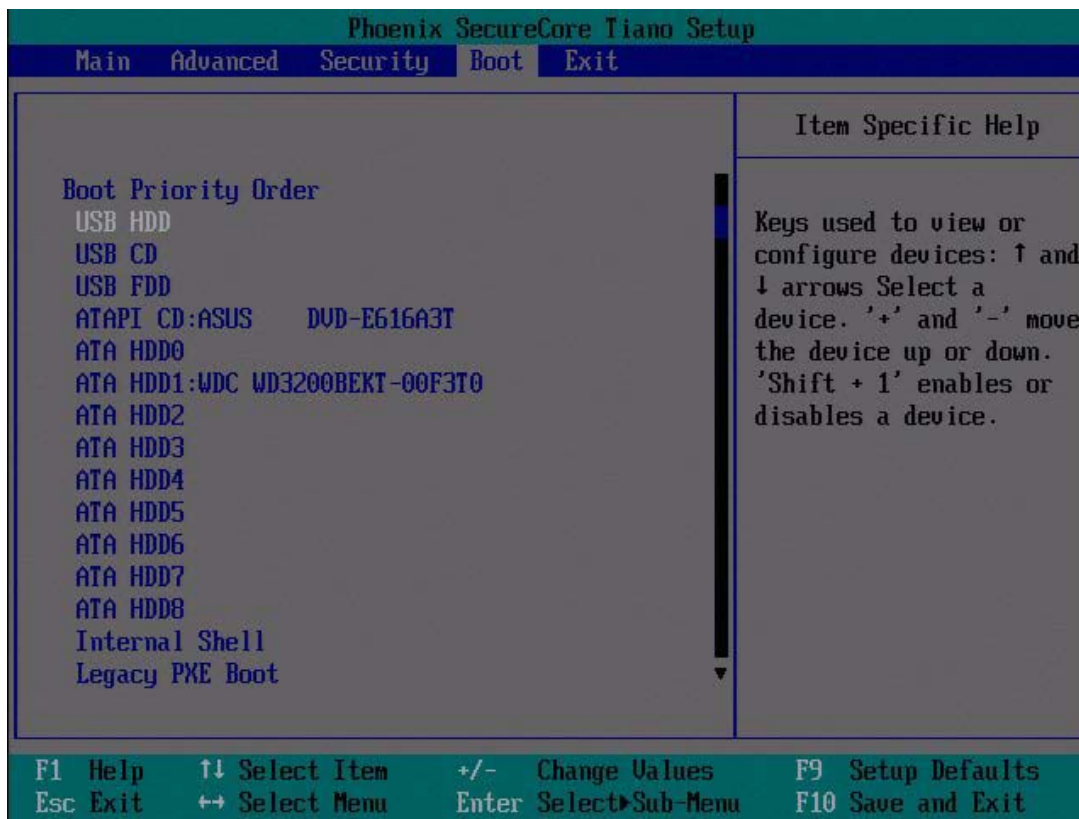


Figure 4.1. System Setup Utility's Boot Menu showing enabled USB drive.

4.2 Legacy USB Keyboard and Mouse

The system supports USB keyboards with or without integrated hubs. The devices must be connected either directly to the platform's USB ports or through a USB hub prior to power-up in order to make sure they are detected and configured properly. The USB keyboard is also available during POST like a traditional PS/2 device so it can be utilized for BIOS configuration and with the integrated debugger.

USB mice must also be connected to the TARGET prior to power-up so they too can be properly detected and configured during POST. The operating system level driver, for example MOUSE.COM in MS-DOS, will need to be installed to enable complete functionality.

WARNING: Both USB and PS/2 devices may be connected to the system simultaneously; however, due to hardware limitations inherent in the PC keyboard controller architecture, it is possible that when using the USB devices at the same time as the PS/2 devices, the keyboard controller may confuse the data streams and intermix them inappropriately, causing the mouse buttons and mouse movements to become erratic. It is advisable to not actively use PS/2 and USB devices at the very same instant to avoid this limitation.

4.3 USB Mass Storage

If configured by the OEM, the system's USB boot feature enables the user to boot the system from a variety of USB storage devices that use no emulation, floppy disk emulation, and hard disk emulation. Some of the USB devices supported include:

- Hard disk drives
- Floppy disk drives
- CD/DVD ROM drives
- CompactFlash readers
- DiskOnKey or memory stick devices
- Zip disk drives

Many of these devices are not configured from the factory to be used as boot devices, therefore it may be necessary to run FDISK /MBR on the device to rewrite the MBR and make it bootable. Any device with a capacity greater than 1.44MB, other than CD/DVD ROM drives, will be automatically treated as a hard disk drive.

4.3.1 Adding USB Drives to the BBS Boot List

The Boot menu enables the user to select which drives will be supported by the system, and also what order they will be scanned to find bootable operating systems. To boot from a USB device, place it early in the system's boot device prioritization list. If a bootable device precedes the USB device, it will still be accessible by the OS booted from the drive that appears earlier on in the list; allowing the USB device to be a data drive or a backup OS boot drive.

When a USB drive is first configured in the BBS list, it will appear with a generic name such as "USB Hard Drive" or "USB CDROM Drive", etc. If, after saving and rebooting the target, the user enters setup again, the system will have automatically scanned the specified generic drive type and obtained the actual name of the drive, such as "Corsair 1GB USB stick SerNo D2961F42R-X", etc.

CHAPTER 5

SECURITY

5.1 Overview and Security Model

SecureCore Tiano has a unified user-level security system. This system provides for authenticating users of the system, and verifying that an authenticated user can perform a firmware-level function, before that function is invoked.

Other types of security, such as chain-of-trust from the first instruction to running the user application, are not a subject of this chapter; this is addressed by the OEM in the BIOS adaptation process.

The purpose of this chapter is to present the threats being addressed by the user-level security feature, and to provide a high-level view of operation of the system, so that the user may understand how the OEM's adaptation of the BIOS fits within the overall capabilities of the system. User-level security in SecureCore Tiano can take on many visible or invisible forms; therefore, this chapter will not offer lots of detail about which permissions are needed to perform which functions in the BIOS, or how users log-in, beyond the most basic display. This is simply because the degree to which OEMs configure this feature is so high, any display shown here would be wrong.

5.1.1 Threat Model

Because SecureCore Tiano is deployed in all types of computing gear, from palm-sized PCs to the largest data switches, industrial controllers and medical equipment, gaming and voting machines, and much more, its primary function is to provide the infrastructure for OEMs to add security to fit the application. Specific threat profiles are not considered in the core firmware itself; these are considered in OEM configurations and customized extensions within the security architecture.

From a system owner's perspective, the purpose of user level security is to prevent unauthorized use of firmware features, including things like accessing the System Setup Utility to change or view system settings or boot from devices.

5.1.1.1 Headless Equipment with Physical Security

Consider a headless embedded system such as a gaming machine, voting machine, or network appliance (router). Physical security is used to prevent tampering with the machine to change its operation, etc. And, these systems do not provide an operator console for the main user. However, they may be connected to the network and offer Telnet or HTTP management capabilities, or they may admit management through a Firmbase console connected to a serial port.

In the network administration and serial port cases, a group of users can gain physical access to the machine and begin working with the system through the interface. Therefore in these scenarios, the system's firmware security provides the solution that identifies particular users and what their exact capabilities are.

5.1.1.2 Mobile (Laptops, Palmtops, and Notebooks)

When equipment is mobile and can operate unplugged, then firmware security plays a different role; it is the first line of defense, not the last one as in the previous category. Commonly, notebooks come with a biometric fingerprint reader that can be used to verify that the owner or authorized users of the device can even turn on the device.

The major threats in the mobile market are theft of the device and access to its data stored on the hard drive. The device can simply be reused by the thief or even sold or repurposed. With sufficient time available, theft of the hard drive data is straightforward without OS-level support; Windows Vista addresses this issue.

In mobile systems then, SecureCore Tiano user security's role is prevention of the device from being useful to anyone except the owner or other authorized persons. The OEM can configure SecureCore Tiano during the adaptation to use biometric readings instead of usernames and passwords, or text-based challenges, providing a deterrent to theft, and making the device effectively useless to thieves.

5.1.1.3 Desktop Systems

Desktop IT systems largely rely on physical security to prevent unauthorized use or theft; for example, desktops are located in offices that may be locked individually, and servers are often located in a secure data room.

However, desktop machines are becoming more and more portable, and in shared workspace environments, it is all too easy for a thief to take a desktop machine in someone's cube and use it at home, wiping the hard drive clean and starting fresh.

Similarly, desktop machines in open work environments make it very easy for coworkers to learn each other's usernames and passwords, so that unauthorized access during non-work hours is possible, compromising data.

The major threats to desktop machines then, are theft and unauthorized access. These threats can be mitigated by using biometrics as in the mobile device class scenario.

5.1.1.4 Server Systems

Server IT systems largely rely on physical security to prevent unauthorized use or theft; for example, they are normally located in a secure data room or bolted into a rack.

Certainly server systems can contain the operational lifeblood of a business—its customer database, source code, and other key information used day-to-day. Therefore, loss, theft of data, or even theft of the whole server, are catastrophic events for different reasons.

SecureCore Tiano user-level security can't address theft of the server unless non-standard hardware is used to tie the server to a specific mode of operation (RFID proximity detector used to start the system, or connection to a security server to verify the server is operating within a secure framework.) Nor can user-level security safeguard access to data stored on hard disks (this is the responsibility of the OS.) But user-level security can address remote configuration at the firmware level; this is commonly supported by servers in the same way that was outlined for network appliances, earlier.

However, desktop machines are becoming more and more portable, and in shared workspace environments, it is all too easy for a thief to take a desktop machine in someone's cube and use it at home, wiping the hard drive clean and starting fresh.

5.2 Basic Username and Password Support

SCT supports (when configured by the OEM) user logins with usernames and passwords, either dynamically as a protected service is requested or in advanced setting.

Once logged in, there are no timeouts implemented by the BIOS firmware; it is up to the Firmware TCB to determine if and when a user's credentials have expired. Every time the BIOS is about to attempt a protected activity, it always passes the currently-logged-in user credentials to the Firmware TCB, and the TCB decides whether the activity is permitted. If no user is logged in and a protected activity is attempted (i.e., boot from a special device, or invoke the debugger), then the BIOS prompts the user to enter credentials first. Please note that this prompting and gathering of user credentials is reconfigurable by the OEM and may involve making a noise to prompt the user to scan a fingerprint, etc.

5.3 Security Controls in System Setup Utility

The System Setup Utility contains a Security menu that can be used to select which activities in the BIOS firmware are protected by the TCB. Security for each item is simply enabled or disabled. Internally, activities at the firmware level in the BIOS that offer protection verify that their protection is enabled, and if so, call the TCB with a special identifier to determine if the currently logged-in user can perform that function, before proceeding.

Chapter 3 presents all of the System Setup Utility's menus, including the Security menu.

Please note that the Security menu may not appear if it is not configured by the OEM. It may also not appear if it is protected, and if a user is not logged in who has permission to view the menu.

In the System Setup, not only are individual menus protected with user-level security controls, but each field is individually protected. This protection can be configured by the OEM; therefore, specific advice about which fields are protected cannot be offered here.